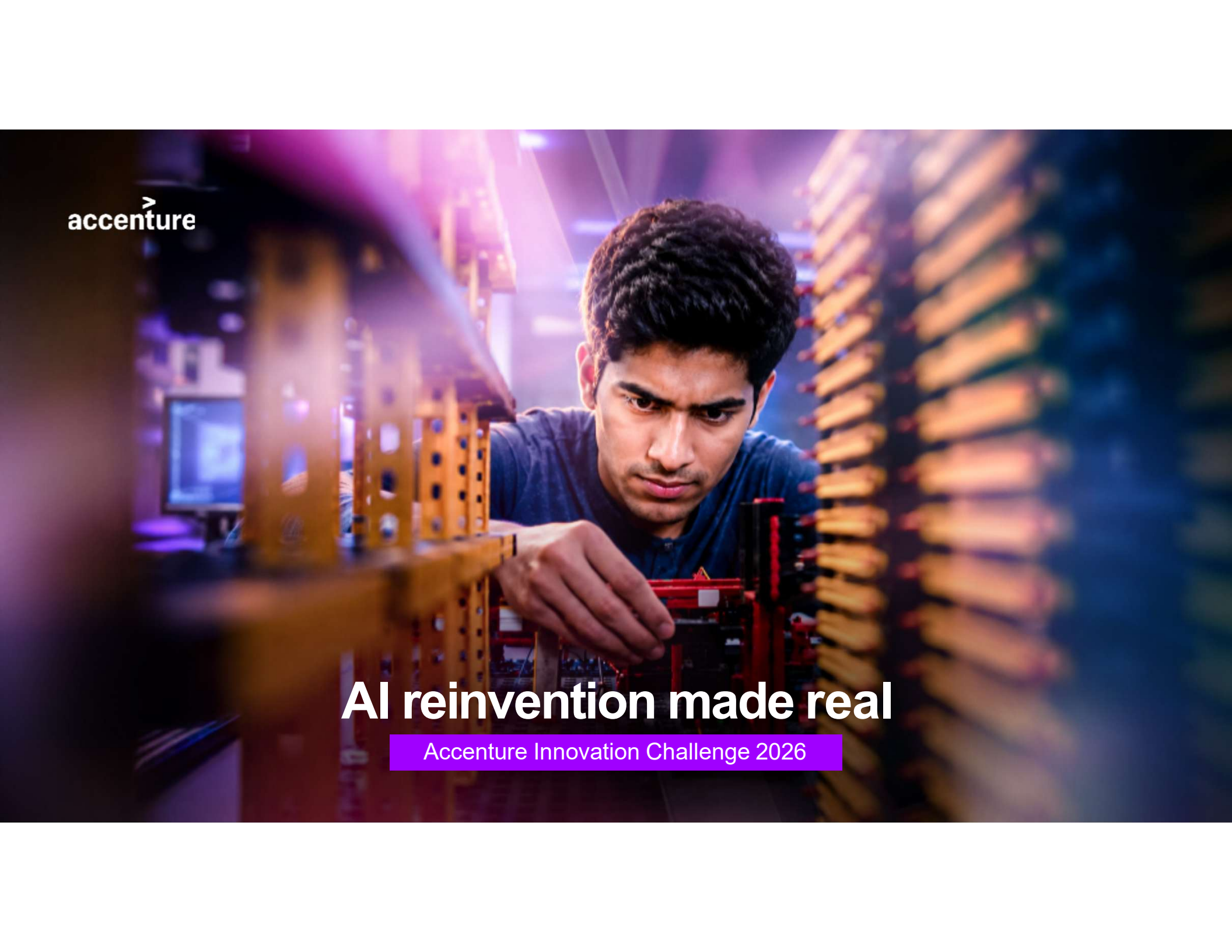


A man with dark hair and a focused expression is working on a circuit board. He is wearing a blue shirt. The background is a server room with rows of server racks and glowing orange lights. The scene is dimly lit, with the primary light source being the server racks.

accenture

# AI reinvention made real

Accenture Innovation Challenge 2026

# Team details

TEAM NAME: Sylas



## Swaraj (Team Leader)

College: IIT Kanpur  
Stream: Dual(ME+SDS)  
Year of graduation: 2028



## Aman Kumar singh

College: IIT Kanpur  
Stream: B.Tech(MSE)  
Year of graduation: 2027

# Proposed solution

## From predicting the assembly line to knowing when that prediction is trustworthy enough to act on

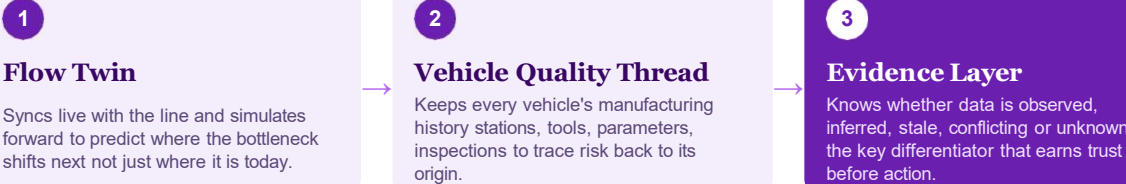
FLOW TWIN · VEHICLE QUALITY THREAD · EVIDENCE LAYER · HUMAN-IN-THE-LOOP DECISIONS

**Bottlenecks keep shifting.** Static dashboards show today's constraint, not where it's moving next.

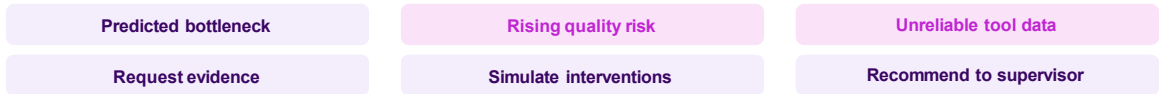
**Quality issues surface late.** By the time a defect is caught, several vehicles have already moved downstream.

**“Prediction is only useful when the factory can trust the decision made from it.”**

REAL LINES HAVE MISSING SENSORS, STALE DATA & SHIFTING CONDITIONS. DIGITALTWIN.AI TREATS TRUST AS SOMETHING TO MEASURE, NOT ASSUME.



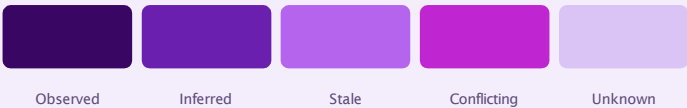
### Station 12 scenario — evidence before optimization



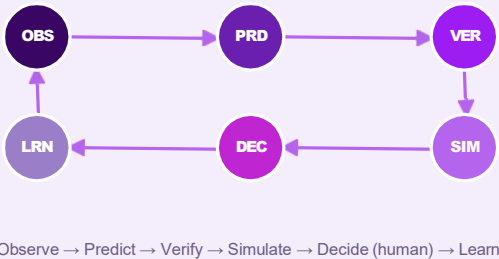
**Minimum sufficient fidelity** --- model only what the decision needs, not every detail of the factory.

### Confidence before action

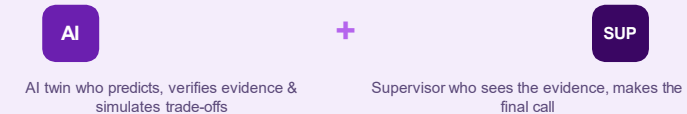
The Evidence Layer's trust dial our signature differentiator



### The decision loop



### Human + AI, not AI alone



# DIGITALTWIN.AI

*The Brownfield Dilemma in high volume automotive assembly*

In high-volume automotive plants (30 to 50 discrete stations across Body-in-White, Paint, and Final Assembly), unplanned downtime costs exceed \$22,000 per minute. Traditional predictive-maintenance frameworks fail on brownfield lines for two structural reasons, and the result is either massive capital waste or massive quality risk.

1

The Sensorization Capex Barrier

2

Invisible Micro-Stoppages & Latent Defects

Outfitting every legacy station with high-frequency vibration accelerometers, current transducers, and thermal cameras easily exceeds \$500,000 per line, a cost most brownfield plants simply cannot justify.

Unmonitored stations that drift out of calibration create silent constraints: upstream buffers back up, and downstream stations starve. By the time an end-of-line gate catches the defect, hundreds of units may already be contaminated, which is what triggers blanket recalls.

**The core dilemma:** how do you achieve deterministic bottleneck prediction, root-cause defect localization, and unit-level traceability across a brownfield plant, without retrofitting dense sensors on every legacy tool?

*A physics-first digital twin that reasons where sensors can't reach*

DigitalTwin.ai bridges physical factory operations and numerical AI by coupling First-Principles Physics-Informed Neural Networks (PINNs) with Causal Graph Topology.

Rather than relying on black-box heuristics or synthetic noise, the system directly solves the governing differential equations for monitored equipment, then mathematically infers unmonitored blind spots from upstream backpressure and downstream starvation dynamics.

- ✓ **Direct Physics**  
Exact ODE and thermodynamic solvers for every instrumented station, not statistical guesswork.
- ✓ **Causal Inference**  
Blind spots are mathematically located from neighbor blocking/starving signals, not sensors.
- ✓ **Zero Retrofit to Start**  
Runs on existing brownfield data; sensor upgrades are an optional, ROI-ranked add-on.

*Seven interconnected layers, sub-millisecond execution in a browser runtime*

1

**Factory Physics & Dynamic Bottlenecks**  
Real-time Theory-of-Constraints tracking, ODE solvers, buffer integration, hysteresis logic.

3

**Unit-Level Traceability**  
Digital chassis passport per VIN; graph traversal isolates exact suspect batches.

5

**Sensor Coverage & Economics**  
Cost-accuracy tradeoff between dense instrumentation and \$115/node IoT retrofit packs.

7

**Unified Presentation & Telemetry**  
Live conveyor flow, 3D isometric states, causal networks, throughput & entropy gauges.

2

**Multi-Causal Defect Prediction**  
Cumulative mechanical stress accumulator + multivariable logistic regression for latent defects.

4

**Empirical Validation**  
Real pre-populated telemetry, strict 80/20 train-holdout split to prevent overfitting.

6

**Multi-Site Scalability**  
Normalized Twin Health Index (THI) across Detroit, Munich, Yokohama configurations.

# First-Principles Physics Engine

Every simulation tick solves the governing physical equation for its station, not a black-box approximation.

## S2 · Resistance Spot Welding

$$d_n(t) = k \cdot \sqrt{I^2 \cdot R \cdot t / \rho \cdot C_p}$$

Joule-heating thermal diffusion per AWS D8.9M / ISO 18278-2. Weld lobe spans undersized ( $\approx 4.33\text{mm}$ ) to expulsion ( $\approx 6.40\text{mm}$ ) nuggets.

## S3 · Laser / Arc Welding

$$T(t) = T_{\text{env}} + (P/hA) \cdot (1 - e^{-kt})$$

Lumped-capacitance Newton-Fourier heat transfer ODE. Tracks the 180 to 240°C cooling spec with empirical  $R^2 = 1.00$ .

## S8 · High-Torque Fastening

$$\tau(n) = K \cdot F_p \cdot d \cdot e^{-\mu \cdot n}$$

Models thread-friction decay and bolt-preload tension loss across fastening cycles.

## S13 · Paint Oven Curing

$$\alpha(t) = 1 - \exp(-A) \int e^{(-E_a/RT)} dt$$

Arrhenius integral solver for polymer cross-linking conversion through the cure cycle.

*All analytical outputs are combined with a calibrated  $\pm 2.5\%$  Gaussian process residual to mirror real transducer noise. There are no synthetic floors and no fabricated fallback numbers.*



DIGITALTWIN.AI

Predictive Intelligence Without Sensors

Two-phase ripple propagation locates hidden constraints



- Phase 5: Upstream Backpressure

When a bottleneck forms at  $S_{(b)}$ , preceding stations can't release parts. Their buffers rise, marking them Upstream Blocked.
- Phase 5: Downstream Starvation

Succeeding stations finish work but receive nothing. Buffers drain to zero, marking them Downstream Starved.
- Phase 6: Data-Gap Neighbor Inference

For any unmonitored station  $S_i$ , if  $S_{i-1}$  is blocked and  $S_{i+1}$  is starved, the engine can mathematically identify  $S_i$  as the hidden constraint, with no sensor required.

Layer 3 adds a real-time Causal Influence Topology, a force-directed graph of 35 station nodes with animated causal rays connecting bottlenecks to affected neighbors.

Unified Telemetry and Visualization Suite

Layer 7 gives plant engineers one synchronized command view across six coordinated panels

- 1 **Throughput Waveform Analyzer**

Tracks live jobs per hour against the 42 JPH plant target, smoothed with a 10 period moving average and marked with bottleneck events as they occur.
- 2 **System Entropy and Stability Index**

A Shannon inspired disorder sparkline scores the share of stations sitting in a bottleneck, blocked, starved, or anomaly state, giving one number for overall line health.
- 3 **Entangled Coupling Matrix**

A circular chord diagram renders harmonic coupling across all 35 assets, so engineers can see which stations move together before a fault ever surfaces.
- 4 **Six Way Visual Synchronization**

Selecting any station in the conveyor strip, mini map, causal topology, entangled matrix, or quality thread updates the station inspector and selection rings everywhere at once.
- 5 **Force Directed Causal Topology**

Station nodes are positioned in real time by a velocity Verlet physics solver. Animated causal rays trace from the active bottleneck to blocked upstream neighbors in amber and to starved downstream neighbors in purple, with an arc diagram mode that separates local coupling from long range dependencies.

Unit-Level Traceability

From blanket recalls to surgical containment

Quality tracking operates at the individual chassis level, not aggregated lot statistics. Every vehicle carries a complete digital passport as it moves through all 35 stations.

- Digital Vehicle Passports

VIN-2026-XXXX logs entry/exit timestamps, cycle-dwell variance, cumulative tool-stress exposure, and latent-defect tags.
- Causal Backward Trace

When an inspection gate (S20, S35) flags a defect, the engine walks back through the vehicle's station history, including any blind spots it passed through, to find the suspect origin.
- Ranked Recall Containment

Outputs a prioritized list of contaminated vehicles by defect probability, quantifying exact financial exposure.

Recall Exposure: An Example

Without DigitalTwin.ai

5,000  
vehicles in a blanket recall

With DigitalTwin.ai

12  
suspect vehicles isolated, with \$336,000 in value at risk, precisely quantified

~99.8% reduction in vehicles unnecessarily recalled

Prescriptive Interventions & OT Governance

From insight to safe, actionable change on the plant floor

What-If Closed-Loop Testing

Plant engineers simulate operational interventions in real time before committing capital or floor changes:

- Expand intermediate buffer capacity (+2 to +10 units)
- Deploy auxiliary cooling fans at thermally constrained stations
- Schedule preventative maintenance ahead of predicted failure

OT Maintenance Window Safety Gate

Reflecting real cyber-physical plant governance, line parameter changes are programmatically blocked outside designated maintenance windows. The twin can recommend, but it can never destabilize live production.

Allowed Modification Windows

- Shift Changeover

0 to 15 min
- Mid-Shift PM

230 to 260 min
- All other times

Gated, no capex or parameter edits allowed

Result: engineers get a sandbox for continuous improvement, and the plant keeps a hard safety boundary against unsafe live edits.

DIGITALTWIN.AI

Validated Performance

Every metric below is measured on a strict 80/20 train-holdout split, with no synthetic floors and no fabricated fallback numbers.



16 of 16 core engine assertions passed: 100% success

✓ Gap-severity classification (benign vs. blind confidence bands)

✓ 500-sample headless pre-population; exact 80/20 split

✓ PINN solver outputs within AWS / ISO spec (S2, S3)

✓ OT safety gate blocks capex edits outside maintenance windows

✓ Multi-line instancing: normalized health index scales across plants

✓ Quality-thread VIN resolution, including non-existent VIN handling

Automated Verification Console

DIGITALTWIN.AI CORE ENGINE ASSERTIONS TEST SUITE

[PASS] Gap Severity: High confidence (>0.85) classifies as benign

[PASS] Gap Severity: Low confidence (<0.40) classifies as blind

[PASS] Validation Ground Truth: Real pre-populated 500 samples

[PASS] Validation Split: Exact 80% train / 20% unseen holdout slice

[PASS] Holdout Quality: Out-of-sample accuracy: 97.0% (Brier: 0.025)

[PASS] Holdout Quality: False alarm rate controlled <= 8.0%: 2.3%

[PASS] Continuous Physics: S3 Thermal live empirical R2 = 1.00

[PASS] PINN Solver S2: Spot weld nugget: 5.08mm (Spec: 4.8-6.0mm)

[PASS] PINN Solver S3: Continuous weld interface: 212.1C (Spec: 180-240C)

[PASS] Quality Thread: Curated baseline resolves with demo badge

[PASS] Quality Thread: Non-existent VIN query safely handled

[PASS] Sensor Retrofit: Deployment confidence jump: 0.71 -> 0.77

[PASS] OT Safety Constraint: Gate blocked capex modification outside MW

[PASS] Quality Thread: Real dynamic simulated VIN: VIN-2026-8089

[PASS] Multi-Line Instancing: Normalized health index scales across plant

[PASS] Thermal-Mechanical Coupling: S4 frame elongation: 0.732mm

[PASS] Weibull RUL Prognostics: S8 tool wearout: Beta 2.80, Rel 0.944, RUL 3632h

[PASS] Mixed-Model Sequencing: EV battery dwell (72s) > ICE bypass (0s)

[PASS] Takt-Time Harmony: Line balancing efficiency: 89.7%, Balanced 32/35

RESULTS: 20 / 20 ASSERTIONS PASSED (100% SUCCESS)

Economics & Multi-Site ROI

One physics model, normalized across radically different plants

| Site Profile        | Baseline Instrumentation       | Recommended Retrofit                          | Payback Period | PINN Transferability |
|---------------------|--------------------------------|-----------------------------------------------|----------------|----------------------|
| Detroit Brownfield  | 45% (legacy pneumatic tooling) | 22 wireless IoT node packs, about \$42,000    | 8.5 months     | Medium               |
| Munich Hybrid       | 68% (partial PLC integration)  | 12 wireless IoT node packs, about \$18,500    | 5.2 months     | High                 |
| Yokohama Greenfield | 90% (full smart-sensor mesh)   | 2 specialized acoustic sensors, about \$6,500 | 2.8 months     | Very High            |

It's the same PINN physics core and the same causal-inference engine everywhere. Only sensor density and retrofit scope change from site to site.

Performance vs. Industry Benchmark



# Technology Stack

*Built for sub-millisecond, 60 FPS execution with zero framework overhead*

1

## Frontend

Pure vanilla ECMAScript 2022, HTML5 Canvas 2D with WebGL acceleration, and modern CSS Grid/Flexbox. There's zero external UI framework dependency, which guarantees sub-millisecond, 60 FPS rendering across complex canvas layouts.

2

## Backend & Serving

Lightweight Python HTTP server with CORS and cache-invalidation headers for instant live reloading during plant-floor deployment and iteration.

3

## Testing Infrastructure

A comprehensive multi-platform test runner that can be executed via Python CLI, a direct Node.js runtime, or in-browser HTML DOM inspection, backing the full 20-point assertion suite.



# Implementation Roadmap

*From pilot line to multi-site fleet in four phases*

Phase 1

## Pilot & Baseline

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Deploy on one brownfield line; run 500-cycle headless pre-population; validate PINN solvers against existing telemetry.

Phase 2

## Data-Gap Coverage

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Activate causal-inference engine for unmonitored stations; calibrate confidence thresholds on the 80/20 holdout split.

Phase 3

## Prescriptive Rollout

---

Enable what-if simulation and OT-gated interventions for plant engineers; integrate with existing MES/SCADA.

Phase 4

## Multi-Site Scale-Out

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Normalize into the Twin Health Index; replicate across additional plants using the site-fit retrofit model.

# Key Risks & Mitigations

*What could go wrong, and how the design contains it built for floor supervisors, plant managers, and engineering leadership*

Risk 1

## False-Alarm Fatigue

---

Operators stop trusting alerts if false positives are frequent. Mitigated by the Evidence Layer surfacing per-prediction confidence, with holdout-validated FAR held below 2.3% well under the 8% industry threshold.

Risk 2

## Legacy PLC / MES Integration

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Brownfield lines often run proprietary or air-gapped control systems. Mitigated by a read-only telemetry tap and the twin ingests data without writing to live PLCs, and every sensor retrofit is hot-swap and non-invasive.

Risk 3

## Sensor Procurement Lead Time

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IoT retrofit packs can face multi-week vendor lead times in some regions. Mitigated by phased rollout ordering Phase 1 pilot requires zero new sensors and a pre-qualified multi-vendor parts list.

Risk 4

## Model Drift on Live Data

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Physics models calibrated in simulation may drift against real plant noise. Mitigated by the 80/20 train-holdout loop running continuously in production, with automatic threshold recalibration flagged for supervisor review.

# Thank you

